VEDA Technology Internship — Task 22 Report

Image Carousel & Slider Documentation & Verification (Scenic Landscapes Theme)

1. Project Overview

This report documents the design, architecture, and verification of **Task 22 (Image Carousel / Slider)** for the VEDA Technology Web Development Internship. The project delivers an interactive, touch-friendly, and accessible image slider engineered strictly with **HTML5**, **CSS3**, and **Vanilla JavaScript (ES6+)**. Styled with an elegant **Light Pastel Green** palette (#F6FAF7 background, #FFFFFF surface cards, #2D6A4F primary forest green, #D8F3DC / #52B788 accents, and #E2EDE5 borders), the application showcases 6 high-definition nature landscape slides with smooth sliding transitions, infinite looping navigation, automated 4-second rotation with hover/focus pause-and-resume, touch swipe gesture support, keyboard arrows, direct dot indicators, and full WAI-ARIA compliance with zero external dependencies.

2. Technical Architecture & Component Stack

Component	Implementation & Technical Details
Semantic HTML5 & ARIA	Structured with <section role="region" aria-roledescription="carousel">, <article role="group" aria-roledescription="slide">, live regions, and role="tablist" indicators.
Responsive Flex & Viewport	Hardware-accelerated horizontal track utilizing CSS transform: translateX(-N * 100%) with cubic-bezier timing (0.4, 0, 0.2, 1) for fluid 60fps sliding.
High-Definition Media	6 landscape slides (Yosemite, Black Forest, Kauai, Kluane, Redwood, Isle of Skye) with 16:9 ratio, responsive object-fit, and lazy loading.
Infinite Looping Logic	Seamless modular wrap-around navigation: clicking Next on slide 6 advances to slide 1, and clicking Previous on slide 1 wraps to slide 6.
Smart Autoplay Engine	Automated 4000ms rotation interval with automatic pause on mouse hover, focusin, or touch interaction, resuming gracefully on blur or mouseleave.
Touch Swipe Navigation	Pure touch event listeners (touchstart, touchend) computing horizontal delta X (threshold ≥ 50px) for smooth mobile swiping.
Interactive Indicators & Counter	Clickable pagination dots with animated active width expansions, accessible aria-selected states, and live numeric fraction counters.
Keyboard & Focus Accessibility	Full arrow key navigation (← / →), visible high-contrast focus rings, and manual autoplay play/pause control button.
Zero Dependencies	100% vanilla code with no external JS libraries or UI frameworks; production-ready for GitHub Pages deployment.

3. Functional Specifications Breakdown

Requirement	Logic & Implementation Description	Status
1. Multiple Image Slides	Features 6 nature photography slides with captions, tags, and descriptive alt text.	**PASSED**
2. Next & Previous Controls	Previous and Next arrow buttons to navigate slides with modular index looping.	**PASSED**
3. Infinite Loop Navigation	Looping wrap-around navigation so users can continuously browse forward or backward.	**PASSED**
4. Smooth Slide Transitions	CSS transform translation with 0.5s cubic-bezier easing for smooth slide animation.	**PASSED**
5. Automatic Slide Rotation	Automatically transitions to the next slide every 4 seconds (4000ms).	**PASSED**
6. Pause on Hover & Focus	Halts the auto-rotation timer whenever the user hovers over or focuses inside the carousel.	**PASSED**
7. Resume on Mouse Leave	Automatically resumes rotation timer when pointer departs or focus moves out.	**PASSED**

8. Clickable Indicator Dots	Dot indicators dynamically highlight current slide and jump directly to clicked slide.	**PASSED**
9. Timer Reset on Interaction	Manual navigation resets the rotation countdown so slides do not switch prematurely.	**PASSED**
10. Touch Swipe Gestures	Mobile touch gesture support allows left and right swiping to transition slides.	**PASSED**

4. Verification & Testing Matrix

Test Scenario	Input Action	Expected State	Actual State	Status
Next Button Navigation	Click Next button on Slide 1	Slide 2 displayed, indicator 2 active, counter shows 2/6	Smooth transition to Slide 2; state updated	**PASSED**
Previous Button Navigation	Click Prev button on Slide 1	Loops to Slide 6, indicator 6 active, counter shows 6/6	Smooth wrap-around to Slide 6 confirmed	**PASSED**
Forward Infinite Loop	Click Next button on Slide 6	Loops to Slide 1, indicator 1 active, counter shows 1/6	Wrap-around to initial slide verified	**PASSED**
Direct Indicator Click	Click dot indicator 4	Direct jump to Slide 4, timer reset to 4000ms	Slide 4 instantly active; timer refreshed	**PASSED**
Autoplay Rotation	Observe idle carousel for 4s	Slide advances automatically to next slide	Slide rotated at exactly 4s intervals	**PASSED**
Hover Pause Verification	Hover mouse over slide media	Timer paused; active slide remains indefinitely	Autoplay halted during hover duration	**PASSED**
Mouse Leave Resume	Move mouse pointer outside	Timer restarts; advances after next full cycle	Autoplay restarted smoothly without skipping	**PASSED**
Keyboard Navigation	Press ArrowRight / ArrowLeft	Slides advance or retreat accordingly	Keyboard events handled accurately	**PASSED**
Touch Swipe on Mobile	Horizontal swipe left (>50px)	Advances to next slide on mobile touch screen	Touch delta recognized; slide transitioned	**PASSED**

5. Deployment & Repository Structure

• **Directory:** Image-Carousel-Slider/

• **Files:** index.html, style.css, script.js, README.md, images/ (6 slides)

• **GitHub Repository:** https://github.com/dharmarayank946/Image-Carousel-Slider.git

• **GitHub Pages Live Demo:** https://dharmarayank946.github.io/Image-Carousel-Slider/

• **Developer:** Nagalambika D Koralli • VEDA Technology Internship